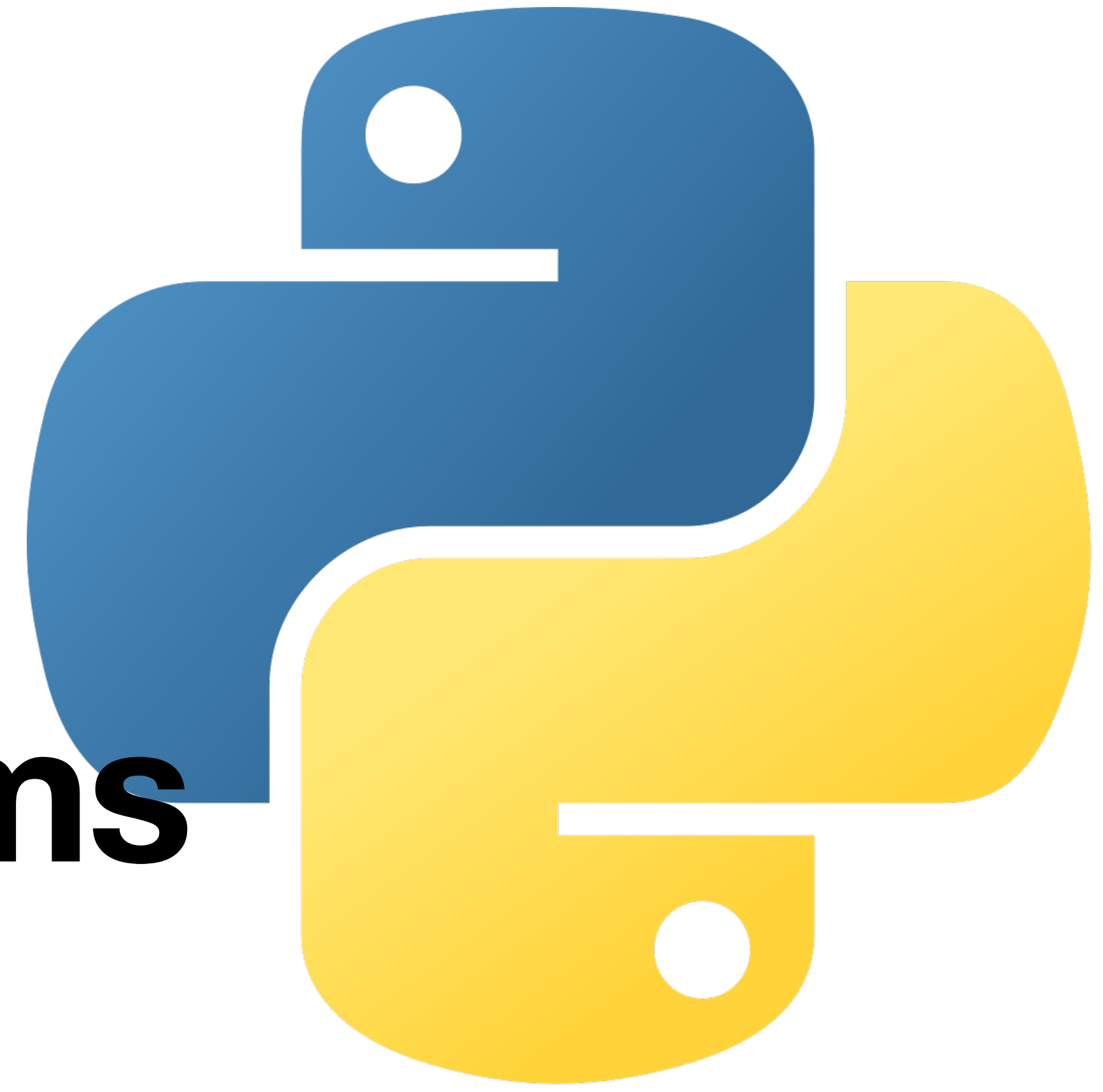


Root Finding Algorithms

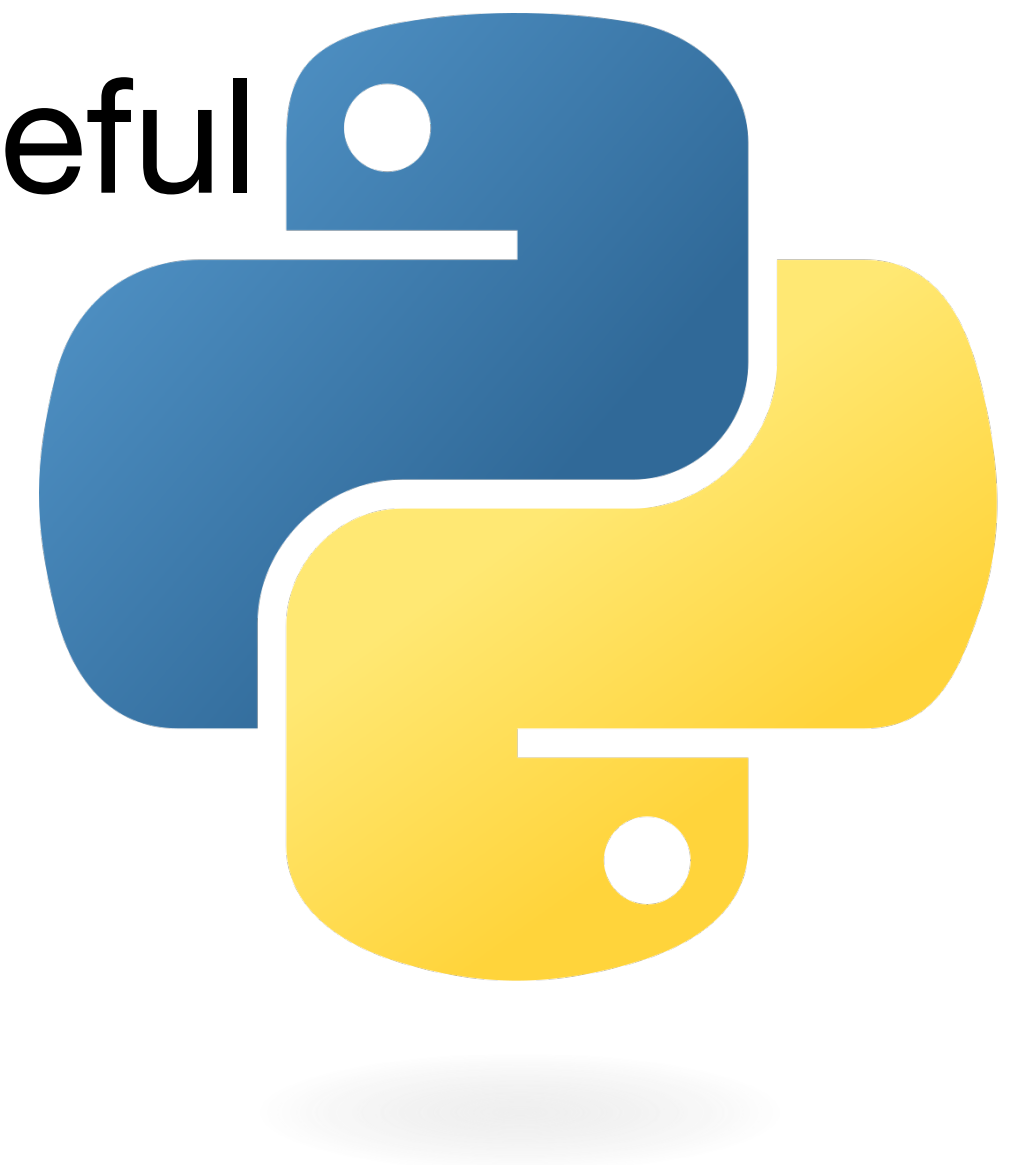
C01 - Winter 2026



Monday, February 9th, 2026

Motivating question

- Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, how can we determine values $x \in \mathbb{R}$ (if they exist), such that $f(x) = 0$?
- What assumptions do we need to make of f ?
- If f was a polynomial, then factoring f gives us useful information about the roots of f .
- However, not all polynomials factor easily,
and not all interesting functions are polynomials.

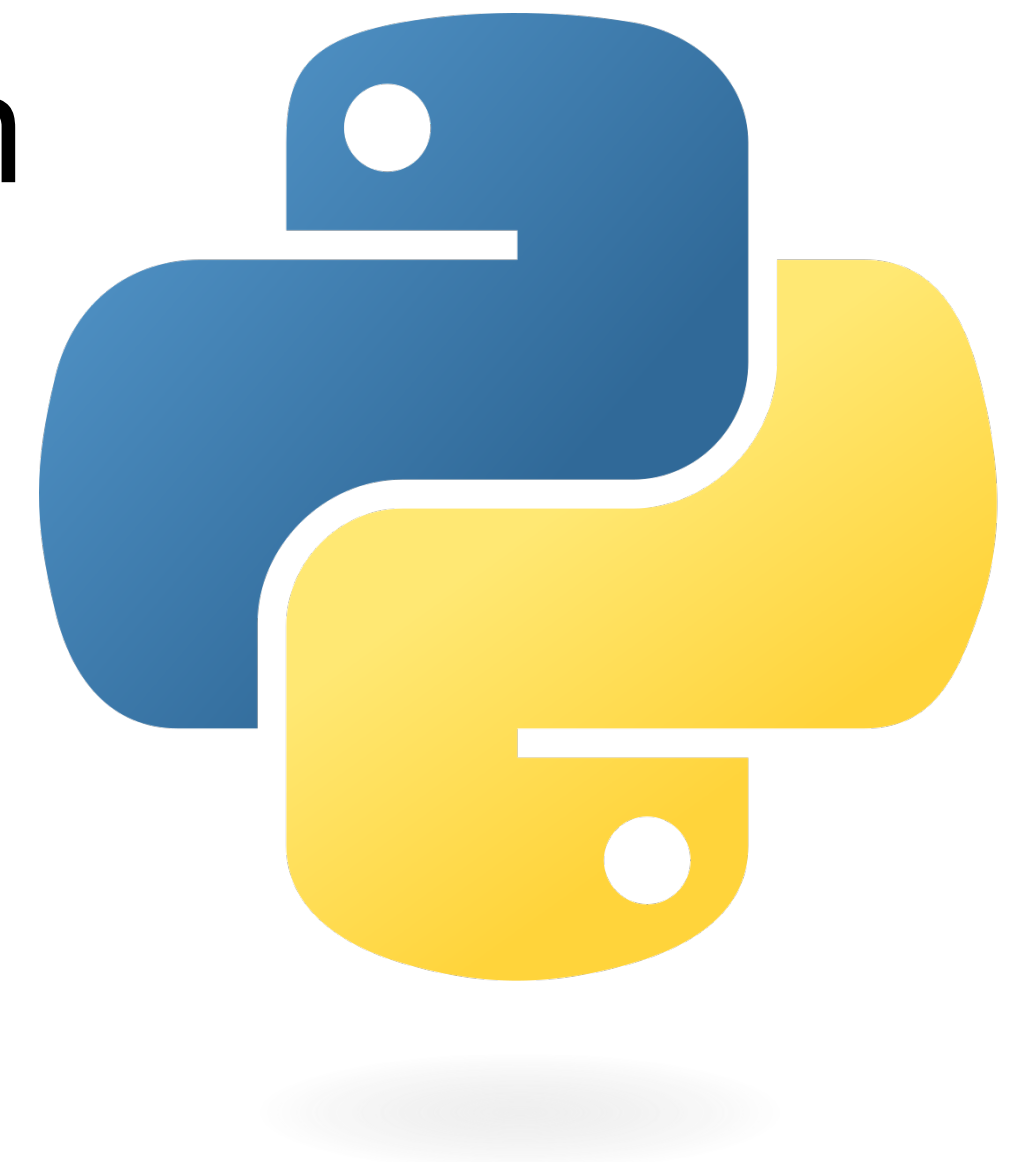


Motivation continued

- For instance, consider the function g as defined below,

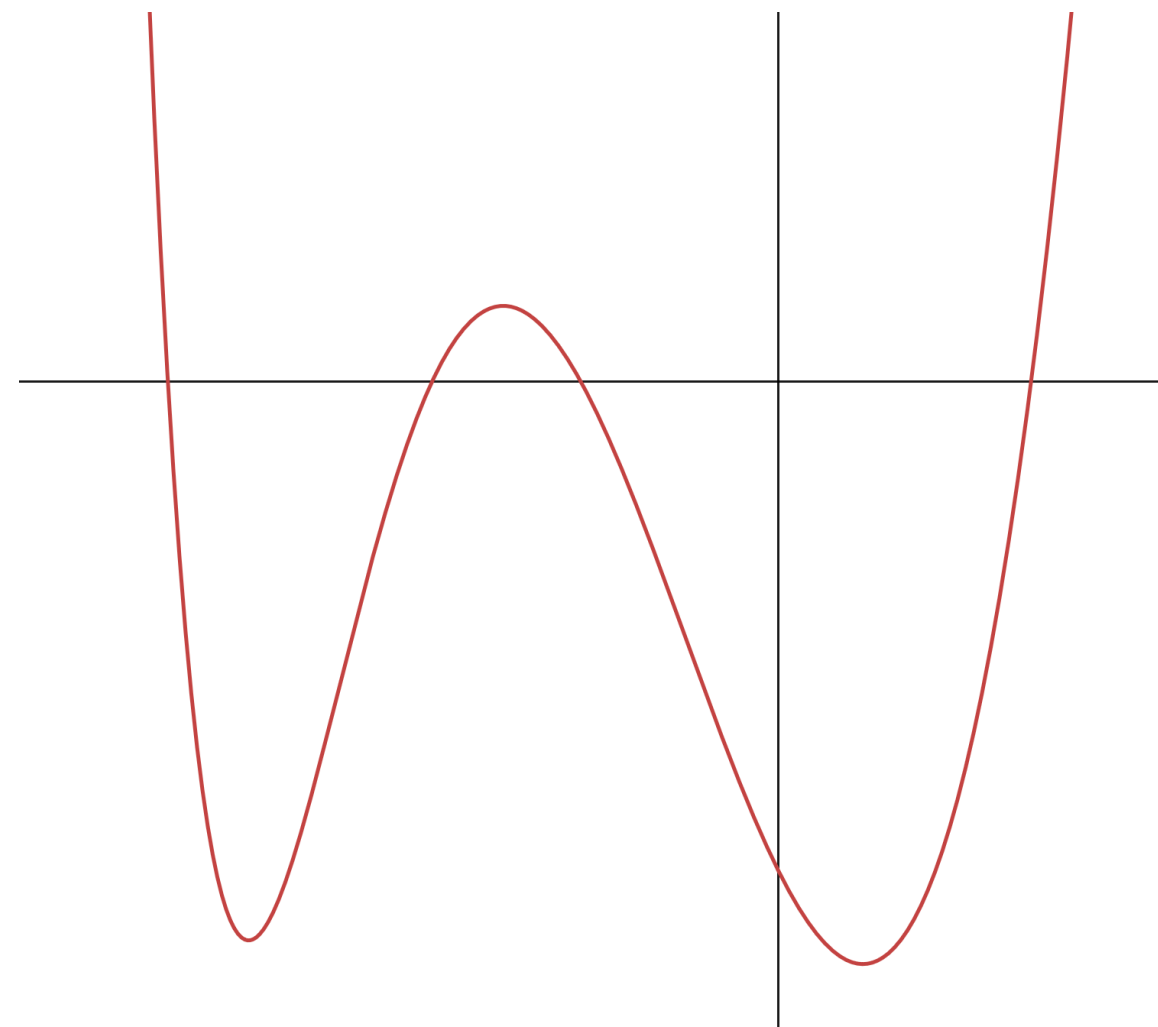
$$g(x) = \begin{cases} 0 & \text{if } x = \pi \\ 2 & \text{else} \end{cases}$$

- It may be very hard to find the roots of a function like g using Python (in fact g has *no roots* in Python - think about this).



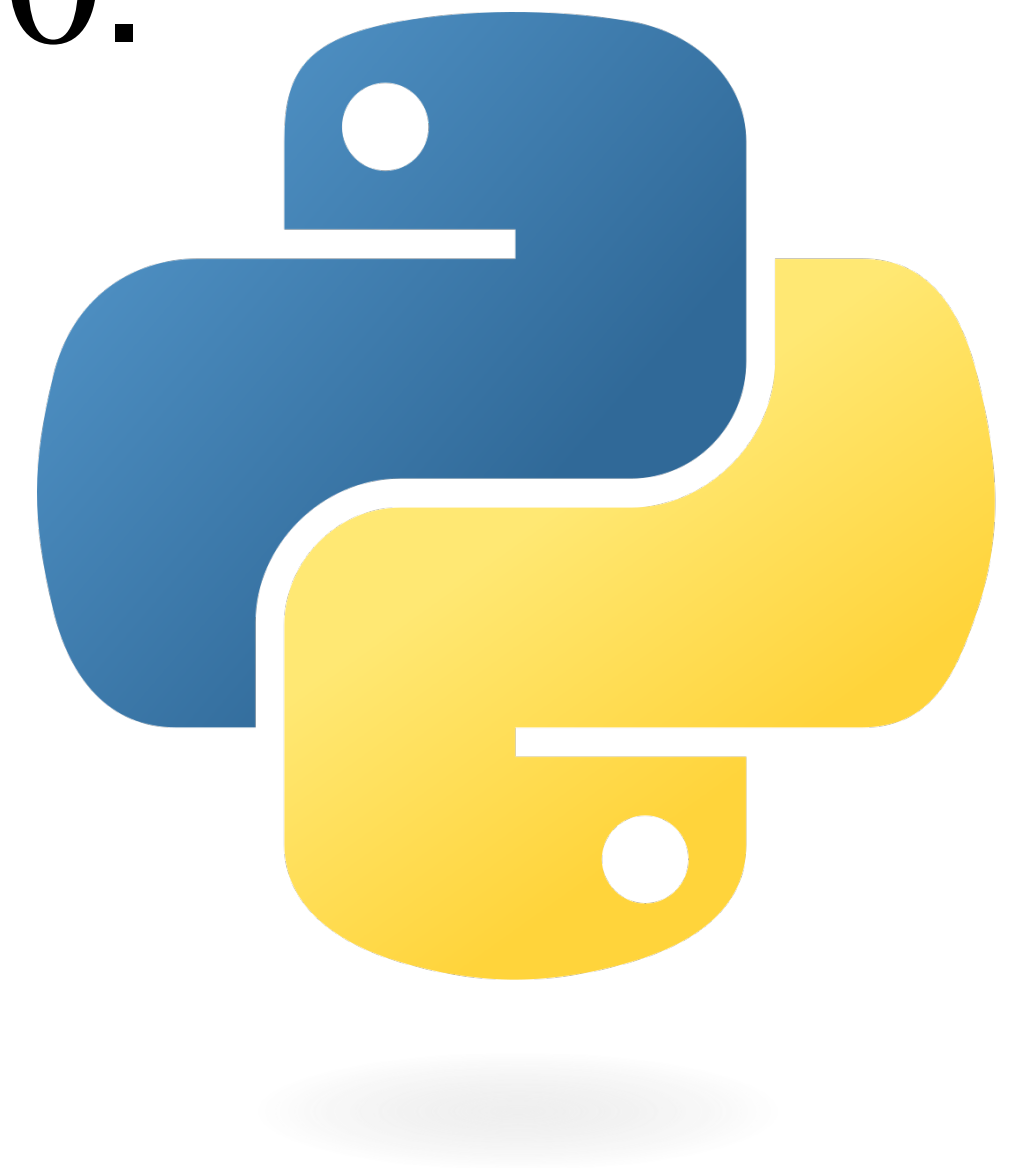
An important tool

- Throwing in the hypothesis that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous gives us a key tool from calculus I. What is it called?
- **Hint:** One can think of a continuous function as a function that can be drawn without lifting the pencil off of the page.



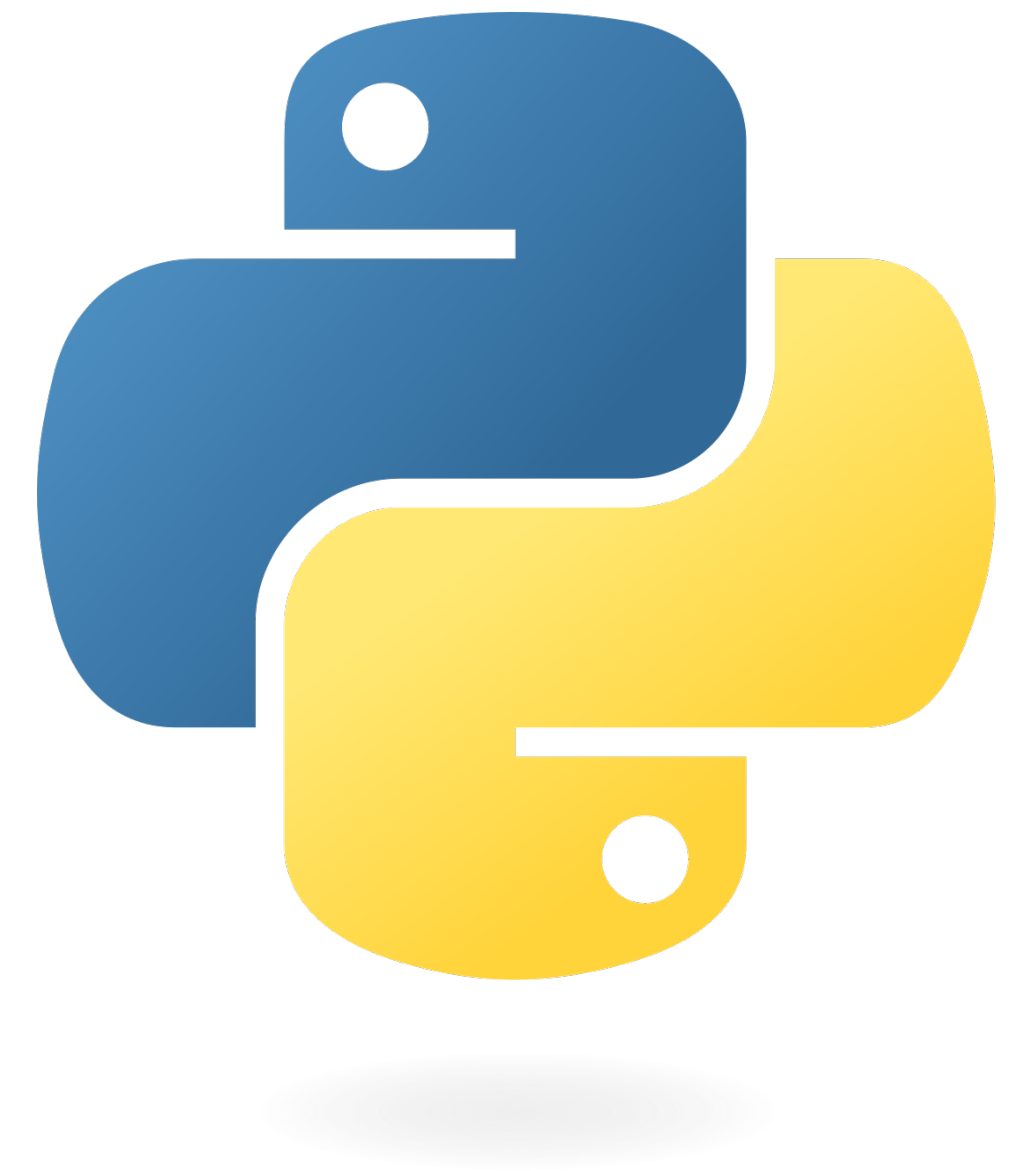
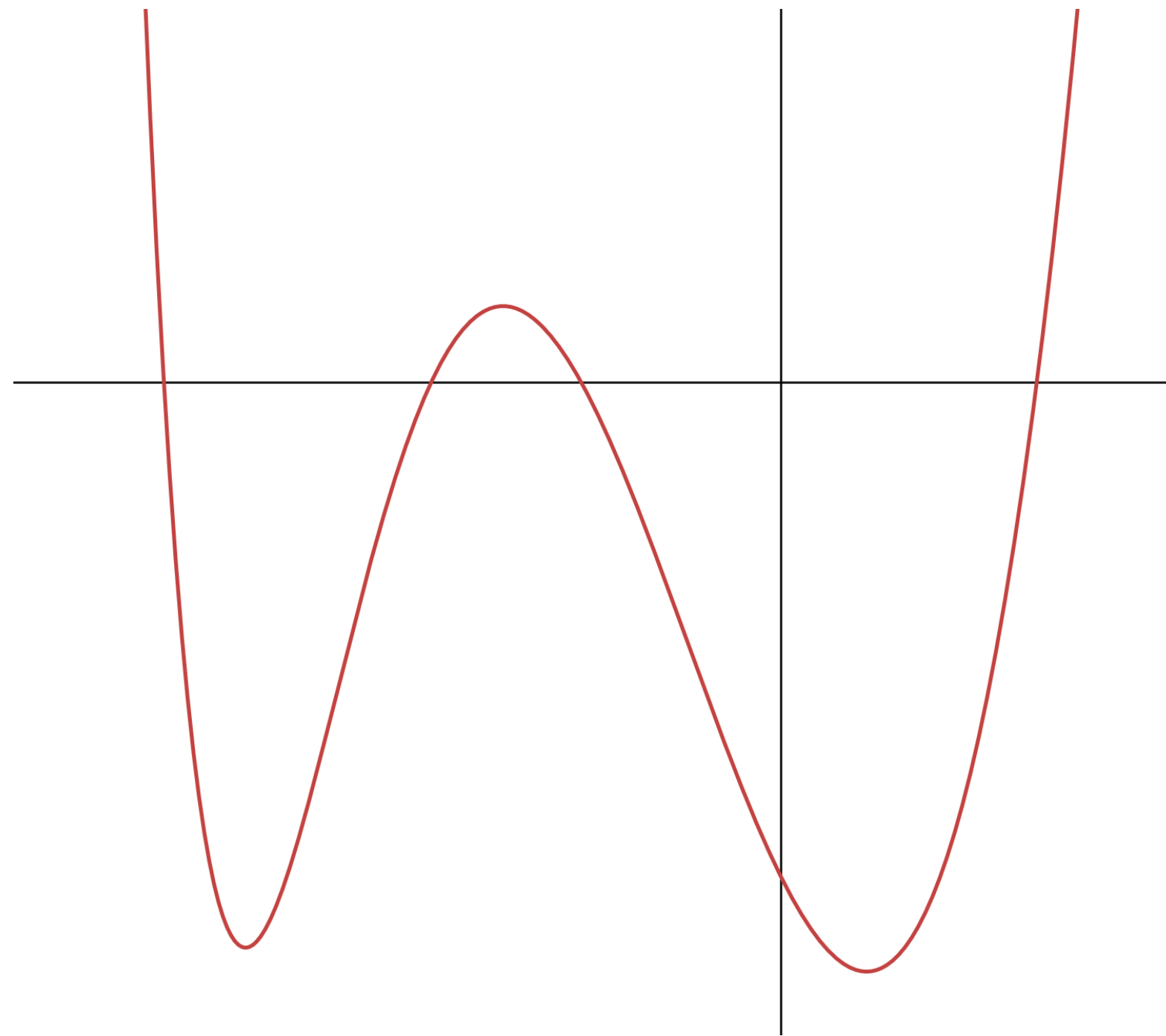
Intermediate value theorem (IVT)

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and let $[a, b] \subseteq \mathbb{R}$. If there exists $c, d \in [a, b]$, such that $f(c) \leq \alpha \leq f(d)$, then there exists $x_0 \in [c, d]$, such that $f(x_0) = \alpha$.
- This theorem is most interesting to us when $\alpha = 0$.



Constructing an algorithm

Question: We have identified IVT as a useful tool, but how can it be leveraged to create an algorithm?



Bisection method - 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and let $[a, b] \subseteq \mathbb{R}$. Suppose that you previously knew (perhaps through the means of graphing f) that f changes sign on $[a, b]$ exactly once.

Let $c = (a + b)/2$ be the midpoint of $[a, b]$.

If $f(a) = 0$, $f(b) = 0$, or $f(c) = 0$, then we are done as we have already found a root of f on $[a, b]$.

Otherwise, we will refine our interval in the following way...

Bisection method - 2

As f changes sign exactly once on $[a, b]$ and as $f(a) \neq 0$, $f(b) \neq 0$, and $f(c) \neq 0$, then f must change sign on $[a, c]$ or $[c, b]$. If f changes sign on $[a, c]$, then $f(a) \cdot f(c) < 0$ and similarly for $[c, b]$.

If $f(a) \cdot f(c) < 0$ update $a = a$ and $b = c$. Otherwise update $a = c$ and $b = b$. Repeat the process until... (we need to choose a stopping condition)

Discussion

Question: Bisection method is a simple and relatively robust root finding algorithm, but it has some key drawbacks. Identify them. (**Hint:** Consider the two images below.)

